

Dataset quality is constrained by table alignment and small sample size despite high image quality.

The dataset contains a total of 38 images across all analyzed files with an average data quality score of 92.4 out of 100 before cleaning and 95.6 out of 100 after. While image quality is high at 99.7 out of 100, the dataset utility is constrained by the table quality score of 91.4 out of 100 and poor cross-type alignment. Issues include 3 table rows referencing missing image files, 4 unreferenced images, and 3 label disagreements between the table and folders. The small overall sample size and mild class imbalance with an imbalance ratio of 1.6 further impact classifier training.

Data quality before	After cleaning	Items	Numbers fact-checked
92.4/100	95.6/100	76 → 75	6

Cross-type findings

CRITICAL Table annotations point to missing files

3 table rows reference image files that are not present in the dataset.

Recommendation: Prioritize locating the missing image files to maintain dataset size; if files cannot be recovered, removing these 3 rows is an acceptable trade-off.

Evidence: mix_links_001

WARNING Unreferenced images exist in the dataset

4 image files are present in the directory structure but have no corresponding entry in the annotations table.

Recommendation: Update the annotations table to include these 4 unreferenced images or remove the files from the dataset.

Evidence: mix_links_001

CRITICAL Label disagreements between table and folders

3 rows in the table contain labels that contradict the folder structure where the images are stored.

Recommendation: Manually verify the ground truth labels for these 3 images and correct either the table entry or the file path.

Evidence: mix_links_001

WARNING Overall dataset quality limited by table alignment

While image quality is high at 99.7 out of 100, the dataset's utility is constrained by the table's lower quality score of 91.4 out of 100 and poor cross-type alignment.

Recommendation: Prioritize fixing the cross-type mapping issues before training, as the high image quality cannot compensate for incorrect labels.

Evidence: mix_overview_001

CRITICAL Duplicate image references in annotations

1 file is referenced by more than one row in the annotations table, which is associated with duplicate training samples or label conflicts.

Recommendation: Review the annotations table to identify and resolve the duplicate entry to ensure each image is uniquely mapped.

Evidence: mix_links_001

Each data type

Each type was profiled, cleaned, analyzed, and reviewed by its own crew. Their full reports are in `cleaned_by_type.zip`.

Table · 1 file(s) · quality 85.1 → 91.4 · rows 38 → 37

- Data Quality Score Improved After Cleaning: The cleaning process improved the data quality score from 85.1 to 91.4 out of 100 by addressing missing values and duplicates.
- Duplicate Rows Removed: The raw dataset contained 1 exact duplicate row, accounting for 2.63% of the data, which was removed during cleaning.
- Missing Values in Confidence and Area: Both confidence and area_px columns had 2.63% missing values with 1 hidden-missing token each in the raw dataset.
- High Rate of Missing Notes: The notes column has a high missing percentage of 44.74% in the raw data and 43.24% in the cleaned data.

Cleaning steps

1. standardize_null_tokens: Changed 1 value in 'confidence', all now missing (e.g. 'N/A' → (missing)); changed 1 value in 'area_px', all now missing (e.g. 'N/A' → (missing)); changed 17 values in 'notes', all now missing (e.g. '' → (missing); 'N/A' → (missing)).
2. parse_percent: Converted 'confidence' to numbers; changed 37 values in 'confidence' (e.g. '70%' → 70; '95%' → 95; '96%' → 96).
3. cast_numeric: Converted 'area_px' to numbers.
4. mark_as_id: Nothing needed changing: the data already met this rule.
5. drop_exact_duplicates: Removed 1 row.
6. add_missing_indicator: Marked 1 row in a new 'confidence_was_missing' column.
7. add_missing_indicator: Marked 1 row in a new 'area_px_was_missing' column.
8. impute_median: Changed 1 value in 'confidence' (e.g. (missing) → 86).
9. impute_median: Changed 1 value in 'area_px' (e.g. (missing) → 4966).
10. flag_outliers: Marked 0 rows in a new 'confidence_outlier' column.
11. flag_outliers: Marked 0 rows in a new 'area_px_outlier' column.
12. standardize_column_names: Nothing needed changing: the data already met this rule.

Image · 38 file(s) · quality 99.7 → 99.7 · images 38 → 38

- Small overall sample size: The dataset contains a total of 38 images across all analyzed files.
- Mild class imbalance present: Circles have 16 images (42.11%), squares have 12 images (31.58%), and triangles have 10 images (26.32%), yielding an imbalance ratio of 1.6.
- Uniform image format and properties: All 38 images are in JPEG format and use RGB mode with 0 corrupt files.
- No image quality defects found: There are 0 tiny, 0 too dark, 0 too bright, 0 near blank, and 0 blurry images out of the analyzed set.
- No duplicates detected: There are 0 exact duplicate groups, 0 near-duplicate groups, and 0 cross-class duplicate files.
- High image data quality score: The image data quality score is 99.7 out of 100 before and after cleaning.
- Vision review notes clean dataset observation: Based on a vision review by gemini-3.5-flash-lite (an AI observation), there are 0 suspected mislabeled items, showing shapes on gradient backgrounds with small dots.
- Cleaning summary shows no dropped files: Cleaning resulted in images changing from 38 to 38, with 0 files removed.

Cleaning steps

1. remove_corrupt: Nothing needed changing: the data already met this rule.
2. drop_exact_duplicates: Nothing needed changing: the data already met this rule.
3. convert_to_rgb: Marked 38 images: to rgb.

Table-to-file links

annotations.csv, column 'image': 35 of 38 rows match a file; 3 point to missing files; 4 of 38 files have no row; column 'label' disagrees with the folder for 3 rows.

Problem	Example
Row names a missing file	circle_100.jpg
Row names a missing file	circle_101.jpg
Row names a missing file	circle_102.jpg
File with no row	shapes_survey/images/circles/circle_003.jpg
File with no row	shapes_survey/images/squares/square_012.jpg
File with no row	shapes_survey/images/triangles/triangle_005.jpg
File with no row	shapes_survey/images/triangles/triangle_010.jpg
Label disagrees (row 4)	shapes_survey/images/circles/circle_004.jpg: table 'square', folder 'circles'
Label disagrees (row 14)	shapes_survey/images/circles/circle_014.jpg: table 'triangle', folder 'circles'
Label disagrees (row 24)	shapes_survey/images/squares/square_008.jpg: table 'circle', folder 'squares'

Next steps

1. Resolve the 3 missing file references and 3 label disagreements between the table and folder structure.
2. Address the 4 unreferenced image files and 1 duplicate file reference in the annotations table.
3. Account for the mild class imbalance across circles, squares, and triangles during model training.
4. Review the remaining missing values and high missing percentage in the notes column.

How the agents checked themselves

Model calls 20 · guardrail catches 4 · self-corrections 5 · review rounds 4 · model switches 7

+4s Triage picked an identifier as the target

target_column 'image' has a different value in almost every row (like an ID or a file name), so a model can't predict it. Pick the column that holds the label or outcome, or use null.

+5s Revised output accepted

+6s Switched models for the strategist

gemini-3.8-flash hit its daily limit; trying the next model in the pool.

+8s Switched models for the strategist

gemini-3.7-flash hit its daily limit; trying the next model in the pool.

+8s Switched models for the strategist

gemini-3.7-flash hit its daily limit; trying the next model in the pool.

+9s Switched models for the strategist

gemini-3.6-flash hit its daily limit; trying the next model in the pool.

+9s Switched models for the strategist

gemini-3.6-flash hit its daily limit; trying the next model in the pool.

+10s Switched models for the strategist

gemini-3.5-flash hit its daily limit; trying the next model in the pool.

+10s Switched models for the strategist

gemini-3.5-flash hit its daily limit; trying the next model in the pool.

+23s Findings failed the fact check

Finding 4 ('Label Class Distribution'): 'label.top.circle' isn't in ['tbl_columns_001', 'tbl_columns_002']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them. Finding 4 ('Label Class Distribution'): 'label.top.square' isn't in ['tbl_columns_001', 'tbl...'

+27s Findings failed the fact check

Finding 3 ('Uniform image resolution and format'): 'widths.min' isn't in ['img_overview_001', 'img_overview_002']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them. Finding 3 ('Uniform image resolution and format'): 'widths.median' isn't in ['img_ov...

+31s Revised output accepted

+37s Reviewer approved the findings

+39s Revised output accepted

... and 5 more in the HTML report.

Cleaning applied

Each data type was cleaned by its own crew.

Charts

Data quality by type

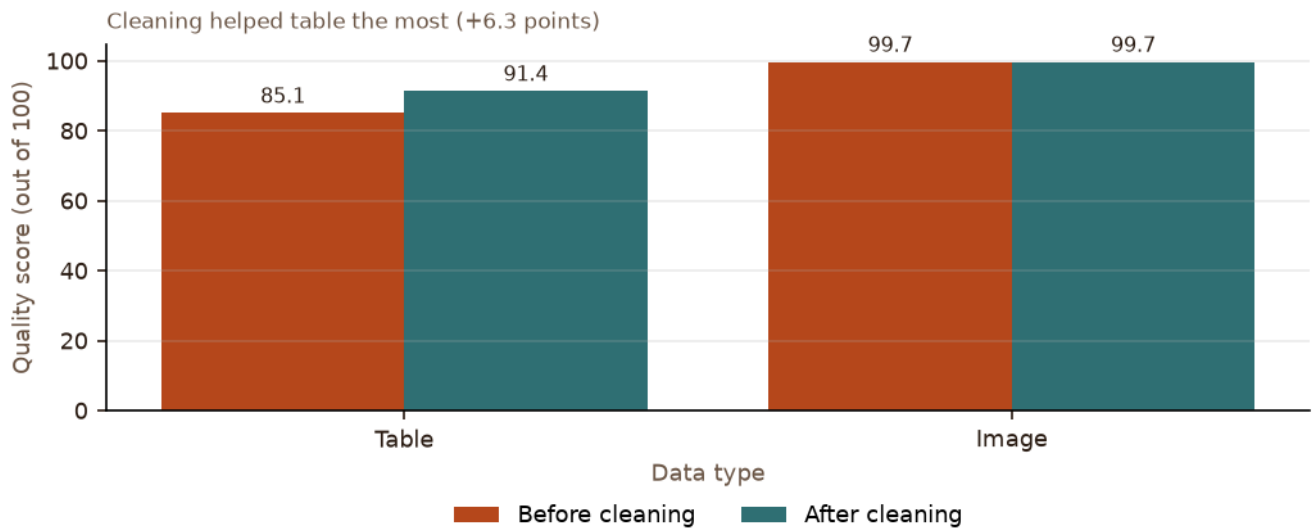


Table: Missing values by column

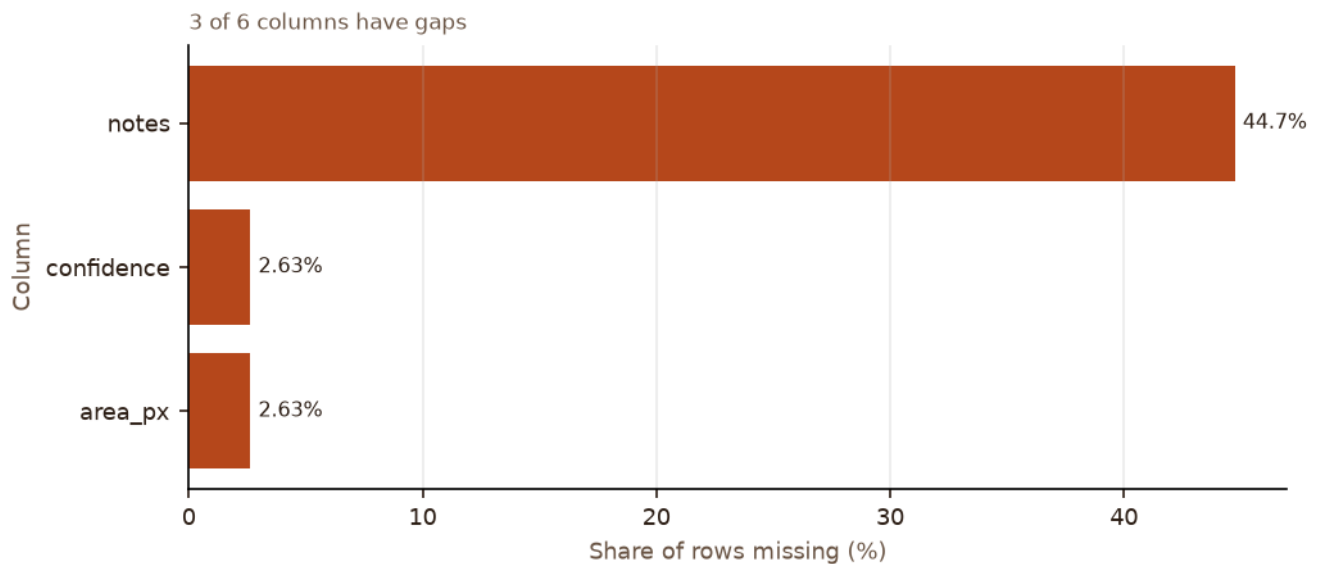


Table: Distribution of confidence

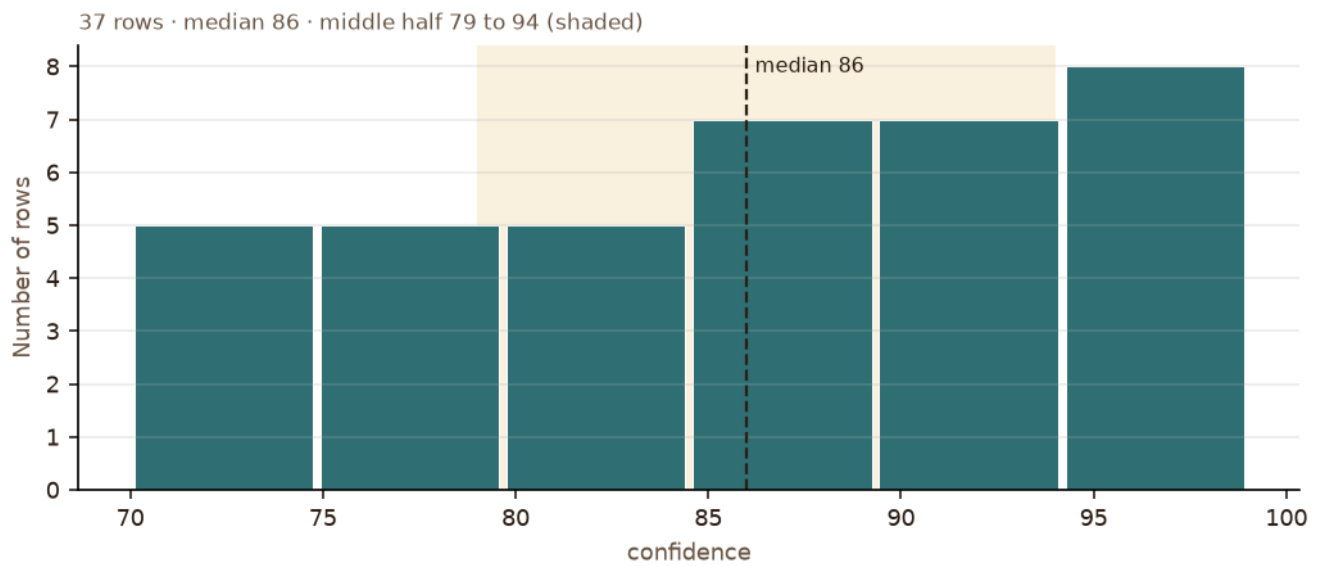


Image: Images per class

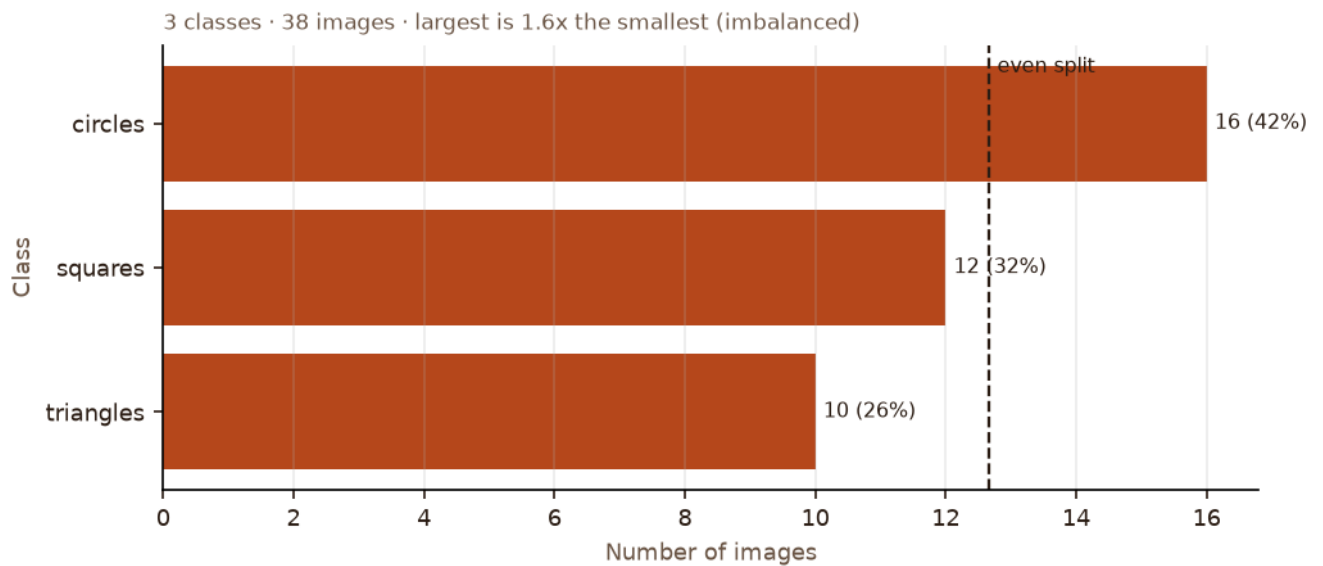
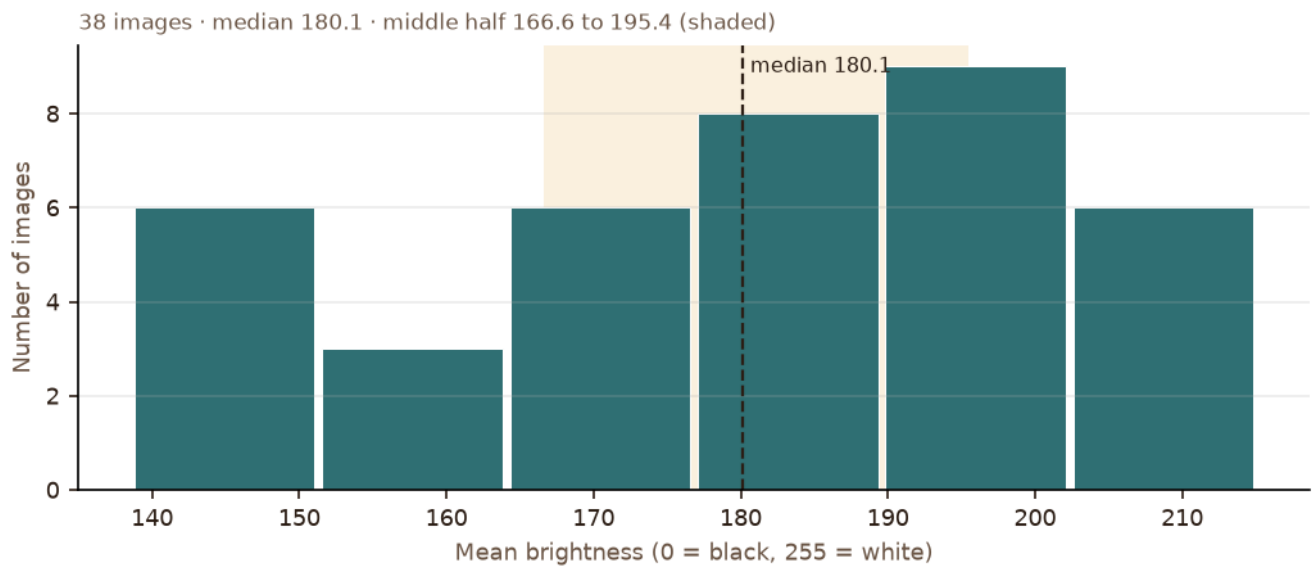


Image: Brightness



Every number in the findings was checked against the evidence artifacts listed. Models used: gemini-3.1-flash-lite, gemini-3.5-flash-lite.